



RETURNSHIELD AI

Return fraud decisioning for modern commerce

PURPOSE-BUILT FOR POST-PURCHASE RISK

Stop
return
fraud
before it
becomes
refund
loss.

Score every return request, explain the decision, and block risky refunds before cash leaves the system.

WHAT IT DELIVERS

0-100

Risk score

3

Decision
bands

Explain

Reason trace

Live

API scoring

Built for the fraud surface checkout tools miss: empty-box claims, wardrobing, item swaps, fraud rings, and suspicious return narratives.

RETURNSHIELD AI

Adopt fast

Score, explain, review, and learn in one control loop.

Merchant value

- Fewer wrongful refunds
- Faster review for good returns
- Better analyst consistency

System output

- Risk score from `0` to `100`
- `AUTO\_APPROVE`, `MANUAL\_REVIEW`, or `HOLD\_REFUND`
- Reason codes and readable explanations

Fast decisions with visible reasoning. Built for merchants who need control, not opaque scores.

OPERATIONAL SNAPSHOT

Score

0-100 risk scale

Explain

reason codes

Learn

feedback loop

HOLD

HIGH-RISK CASES

Product features

ReturnShield AI works as an operational fraud console, not just a scoring endpoint.



Return scoring
Scores every return before the refund is released.

Explainability
Shows why a request was flagged in plain language.

Case review
Routes medium and high-risk returns to analysts.

Feedback loop
Stores analyst decisions for retraining and follow-up.

Fraud ring visibility
Surfaces connected accounts and repeat behavior.

Reason codes
Turns score output into actions the team can trust.

TYPICAL WORKFLOW

1. Return request arrives.
2. Payload is normalized.
3. Rules and models score the case.
4. The return is approved, reviewed, or held.
5. Analyst feedback is stored for improvement.

Algorithm and accuracy

The model combines signals instead of trusting any single field.

LAYER	SIGNAL	ROLE
Rules	Policy checks	Fast guardrails for known patterns
Structured ML	Customer, order, shipment, return fields	Finds hidden tabular patterns
NLP	Return reasons and support text	Catches suspicious language shifts
Anomaly detection	Outliers and unusual combinations	Flags abnormal behavior

```
final_score =
  (rule_score * 0.30) +
  (structured_ml_score * 0.30) +
  (nlp_score * 0.25) +
  (anomaly_score * 0.15)
```

0-39	AUTO_APPROVE	Low risk, release the refund.
40-69	MANUAL_REVIEW	Needs analyst inspection.
70-100	HOLD_REFUND	Strong fraud indicators present.

DECISION LOOP

REQUEST

SCORE

REVIEW

LEARN

ACCURACY STORY

- Precision on obvious fraud
- Recall on subtle fraud
- Consistency across repeated cases
- Reviewability for analysts and auditors

The repo tracks precision, recall, and F1 in training runs, but it does not publish audited benchmark percentages. That keeps the pitch credible.